YEAR 7 · SCIENCE 3.1 · LAB

Drop It!

How fast are you? Do heavy things fall faster?

Name: _____ **Class:** _____ **Date:** _____

My group: 1. _____ 2. _____ 3. _____

Predict first — on your own

Mr Bowen drops a heavy ball and a light ball, the same size, at the same time. Which lands first? ☐ heavy ☐ light ☐ same time

I think … because …

_____

Stay safe

- Only drop objects **Mr Bowen has checked**.
- **Let go. Never throw. Never catch.**
- Part 2: feet on the floor. No chairs or desks.
- Part 3: stay **behind the railing**. Never lean.
- Part 3: drop only after the Ground person shouts **"Clear!"**

Two tools: you can use a calculator

Mean = add up all the values, then divide by how many there are.

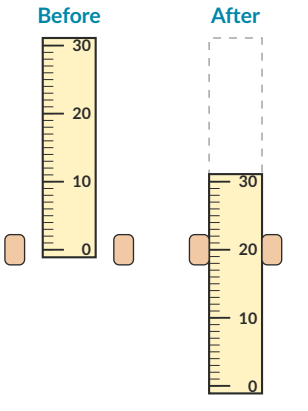
Speed = distance ÷ time, in **m/s** (**metres per second**). **cm to m:** ÷ 100

Mr Bowen: 20 cm = 0.20 m, in 0.20 s, so speed = 0.20 ÷ 0.20 = **1.0 m/s**.

Part 1 · The Ruler Catch — How Fast Are You?

Jobs: Catcher | Dropper | Timer

1. **Catcher:** rest your arm on the desk. Thumb and finger open, at the **0**.
2. **Dropper:** hold the top. Let go with **no countdown**!
3. **Catcher:** catch it. Read the cm at the **top of your thumb**.
4. **Timer:** try to time the fall with a stopwatch.
5. **5 catches each.** Then change jobs.



20 cm = 0.20 s

1A The ruler table: where you caught it → your reaction time

Caught at (cm)	1	2	3	4	5	6	7	8	9	10
Time (s)	0.04	0.06	0.08	0.09	0.10	0.11	0.12	0.13	0.13	0.14
Caught at (cm)	11	12	13	14	15	16	17	18	19	20
Time (s)	0.15	0.15	0.16	0.17	0.17	0.18	0.18	0.19	0.19	0.20
Caught at (cm)	21	22	23	24	25	26	27	28	29	30
Time (s)	0.20	0.21	0.21	0.22	0.22	0.23	0.23	0.24	0.24	0.24

1B My five catches

Catch	1	2	3	4	5	Total	Mean (÷ 5)
Caught at (cm)							
Ruler time (s)							
Stopwatch time (s)							

Speed of the ruler = mean catch (m) ÷ mean time = _____ ÷ _____ = _____ **m/s**

Think Your ruler times and stopwatch times are different. Why is a stopwatch no good for timing the ruler? Use **reaction time**.
The fall only takes about … The Timer …

Part 2 · Two Objects in the Lab — Do Heavy Things Fall Faster?

Our objects, the **same size and shape**: **A** (heavy) _____ **B** (light) _____

Jobs: Dropper | Timer A | Timer B Checked by Mr Bowen ☐

1. Put tape on the wall, as high as the **shortest** person can reach. Drop from this mark **every time**.
2. Dropper: hold A and B at the tape. Say **"3, 2, 1, drop!"** and let go of both together.
3. Timer A times A. Timer B times B. Everyone listens: **one bang or two?**
4. Do **10 drops**. Change jobs after drop 3 and drop 6.

Height of the tape: _____ cm = _____ m

2A Ten drops

Write each time like this: 0.63. Last row: who landed first? **A**, **B** or **S** (same time).

Drop	1	2	3	4	5	6	7	8	9	10	Total	Mean
A, heavy (s)												
B, light (s)												
First? A, B, S												

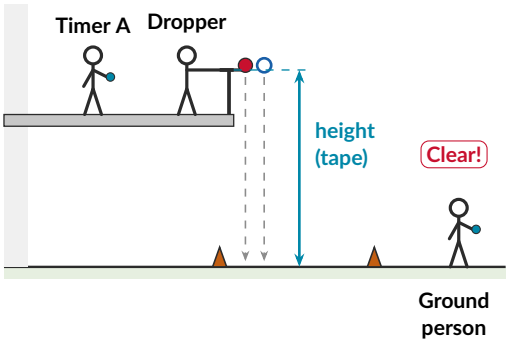
Speed of A = height (m) ÷ mean time = _____ ÷ _____ = _____ **m/s**

Think Why did we **change** Timers? Why must A and B be the same size and shape?
So that a slow Timer … If one is bigger or flatter, the air …

Part 3 · From the Balcony — How Fast Does It Go?

Upstairs: Dropper Timer A
Downstairs: Ground person, who also times B

1. **Height:** the Ground person holds the **0** of the long tape on the ground. Upstairs, read it at the **top of the railing**. Three readings.
2. The Ground person stands well back and shouts **"Clear!"**
3. The Dropper holds A and B out over the railing, level with the top: **"3, 2, 1, drop!"**
4. Timer A times A. The Ground person times B, then **walks** them back up.
5. Change jobs after every drop. Do **6 drops**.



3A Height and prediction

	Reading 1	Reading 2	Reading 3	Mean (÷ 3)
Height (m)				

How long a fall takes with **no air**:

Height (m)	1.8	2.0	2.2	3.0	3.5	4.0	4.5	5.0	5.5	6.0	7.0
Time (s)	0.60	0.63	0.66	0.77	0.84	0.89	0.95	1.00	1.05	1.10	1.18

Predict. From our height, the fall takes about _____ s.

Which lands first from up here? ☐ A ☐ B ☐ same time

3B Six drops from the balcony

Drop	1	2	3	4	5	6	Total	Mean (÷ 6)
A, heavy (s)								
B, light (s)								
First? A, B or S								

Speed of A = height (m) ÷ mean time = _____ ÷ _____ = _____ **m/s**

Think Your reaction time is about 0.2 s. Which part gave the more **accurate** time, Part 2 or Part 3? Why?
The fall in Part … was longer, so 0.2 s is …

Reflection · Dear Aristotle

More than 2000 years ago, **Aristotle** said that heavy things fall faster. People believed him for about 2000 years. Write him a polite letter.

Your letter in four parts

Claim	your answer. Heavy things …
Evidence	at least **two numbers** from your tables. In the lab, the mean times were … From the balcony, …
Reasoning	the science. Gravity pulls the heavy object harder, but it is also harder to speed up, so …
Next time	one way to make the test more accurate. Next time we could …

Dear Aristotle,

From, _____

Teacher Copy

T1 Before the lab

- [ ] **Objects**, one pair per group, the same size and shape: a tennis ball and a ball of plasticine rolled to the same size, or two identical water bottles (one full, one a quarter full, caps taped). Nothing glass, metal or sharp. Initial each pair. A ping-pong ball against a golf ball *will* lag from the balcony (the air) — a good demonstration to do yourself at the end.
- [ ] Per group: a 30 cm ruler, 2 stopwatches (phones are fine if allowed), masking tape, a metre stick, a calculator. A 10 m tape that reaches the ground from the balcony (one can be shared).
- [ ] **The balcony:** measure the height yourself first so you can check 3A. Tell the office and the teachers below. Cone off the landing zone. Be on the balcony yourself. Two groups at a time; the rest finish Part 2.
- [ ] **Rain plan:** a stairwell, or Part 3 as a demonstration.

Timing. About 15 minutes for Part 1, 15 for Part 2, 20 for Part 3, 10 for the letter. If it runs long, the letter is homework.

1 Part 1

Catches land at **10–25 cm**: ruler times **0.14–0.22 s**, means near 0.18–0.20 s. The ruler's speed is **0.7–1.1 m/s**; a quick check is 5× the mean time (0.20 s → 1.0 m/s). Stopwatch times will be all over the place, 0.1–0.5 s, some 0.00. That is the result, not a mistake.

Think: "The fall only takes about 0.2 s, and the Timer also has a reaction time of about 0.2 s, so they start and stop the stopwatch late." Accept anything that blames the Timer's reaction and says the fall is too short.

2 Part 2

The tape mark is about **1.8–2.1 m** (0.60–0.65 s). Stopwatch means 0.5–0.9 s, with A and B within about 0.05 s of each other. Most drops "S"; the rest split roughly evenly. Speed about **3 m/s**.

Think 1: "No. The mean times were 0.64 s and 0.65 s, nearly the same, and they landed at the same time in most drops." A student who says "A was faster by 0.01 s" needs the question *is 0.01 s bigger or smaller than a reaction time?*

Think 2: "So that a slow Timer does not make one object look slower every time. If one is bigger or flatter, the air pushes up on it more and slows it down, so the test is not fair."

3 Part 3

One storey plus a railing is usually **4–5 m**: about **0.9–1.0 s**. The three tape readings should agree within a few cm. Speed of A about **4.5–5 m/s**. Same-size pairs still land together from 5 m.

Think: "The fall in Part 3 was longer, about 1 s instead of 0.6 s, so 0.2 s is a smaller part of it and the time is more accurate." This is the reason the lab goes upstairs; make sure it is said aloud.

R Dear Aristotle

Check in order: a clear claim; two numbers *with units*; the reasoning from 3.1 slide 39 (pulled harder, but harder to speed up). A model:

"Dear Aristotle, heavy things do not fall faster. In our lab the heavy ball took 0.64 s and the light ball 0.65 s. From the balcony they took 0.97 s and 0.98 s and landed together. Gravity pulls the heavy ball harder, but it is also harder to speed up, so they fall together. Sorry!"

More accurate: film it in slow motion and count frames; more drops; drop from higher; one person times every drop. Deadpan humour in the letter is welcome — it makes a good display.